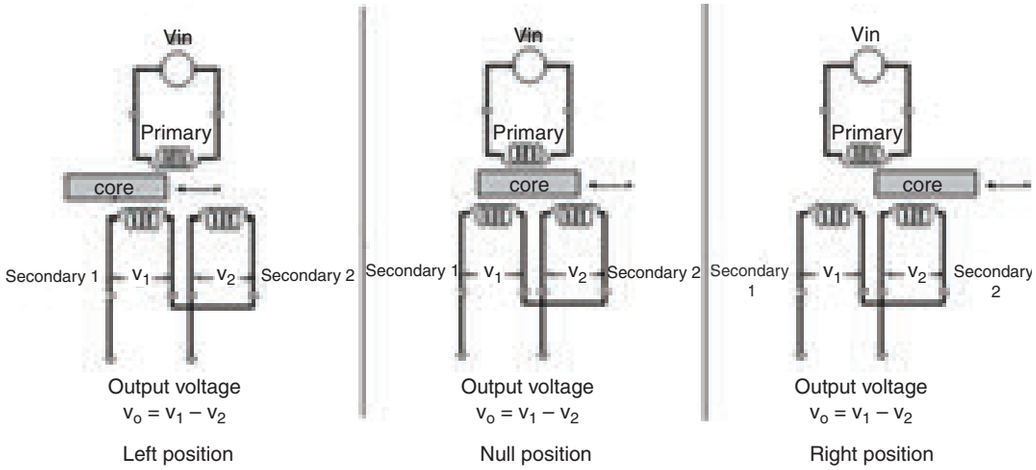


Fig. 7.11 LVD Transformer

to the sensing element of displacement is to be measured. This core is movable with respect to the applied displacement.

When there is no force, the central core occupies null position as shown in figure 7.11. At this time the magnetic coupling of primary to secondary are equal. Hence the two output voltages of secondary windings V_1 and V_2 are equal.

As the core is moved towards the left from its null position, the secondary winding voltage V_1 increases over V_2. This is shown in figure 7.11.

As the core is moved towards the right from its null position, the secondary winding voltage V_2 increases over V_1. This is shown in figure 7.11.

In this way the displacement is measured by LVDT.

c. Advantages of LVDT

i. LVDT has high sensitivity.
ii. Consumes very less power.
iii. Good frequency response.
iv. LVDT is more reliable device.
v. It is very rugged device.

d. Disadvantages

i. Relatively large displacement is needed to make output.
ii. Affected by vibrations.
iii. Temperature affects the performance of LVDT.

7.5.4 Capacitive transducers

The capacitance of a parallel plate is given by,

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

Where,

ϵ_0 = permittivity of free space.
ϵ_r = the relative permittivity of free space
A = Area of plates
d = Distance between the plates.

As the capacitance of a capacitor depends on the above parameters, such transducers can be used to measure the non-electrical quantity such as displacement, force, pressure etc.

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